

Charlie Mpetha

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EMPLOYMENT

NPP Fellow Sep 2025 - Present
NASA Goddard Space Flight Center

EDUCATION

Ph.D in Cosmology Sep 2020 - Apr 2025
Institute for Astronomy, University of Edinburgh

MSc in Astrophysics Sep 2019 - Aug 2020
Astrophysics Research Institute, Liverpool John Moores University

MPhys Oct 2015 - Jun 2019
Department of Physics, Lancaster University

AWARDS AND FUNDING

2023 Leverhulme Study Abroad Scholarship
£23,040 over one year.

2019 Research Academy Prize

2019 Outstanding Final Year Performance

2018 Cormack/Piazzzi Vacation Research Scholarship
£266 per week for six weeks.

PUBLICATIONS

Cosmology from UNIONS weak lensing profiles of galaxy clusters.
Mpetha C. T. et al. 2025, MNRAS, 10.1093/mnras/staf1538

The infall region as a complementary probe to cluster abundance.
Mpetha C. T. et al. 2024, MNRAS, 10.1093/mnras/stae1637

Constraining cosmological parameters using the splashback radius of galaxy clusters
Haggar R., Amoura Y., Mpetha C. T. et al. 2024, ApJ, 10.3847/1538-4357/ad5cee

Impact of weak lensing on bright standard siren analyses
Mpetha C. T. et al. 2024, PRD, 10.1103/PhysRevD.110.023502.

Future prospects of testing extensions to Λ CDM through the weak lensing of gravitational waves

Mpetha C. T. et al. 2023, PRD, 10.1103/PhysRevD.107.103518.

Gravitational redshifting of galaxies in the SPIDERS cluster catalogue

Mpetha C. T. et al. 2021, MNRAS, 10.1093/mnras/stab544.

TEACHING AND MENTORSHIP

Volunteer Tutor

Oct 2024 - Apr 2025

- Volunteer tutoring with the Sudanese Community Group Edinburgh.

Summer Student Mentor

July - August 2024

- Mentored a high school student on a project visualising simulated galaxy clusters.

Teaching Assistant

Oct 2020 - Present

- Courses: Cosmology (MSc), Introductory Astrophysics, Modern Physics, Physics 1A+1B (BSc).

- Nominated for *Student Tutor of the Year* in the 2022 Student-Led Teaching Awards.

Maths and Physics Tutor

Dec 2019 - Sep 2023

- Tutored school students from ages 11 – 18, and adults, on a range of topics in Maths and Physics.

PRESENTATIONS

Cosmological and Astrophysical Synergies: Tactics for the Latest Era

September 2024, Tagliolo Monferrato, Italy.

University of Maryland

Visiting speaker. August 2024, Maryland, USA.

Cosmology from Home

June 2024, Online conference.

UNIONS Collaboration - Invited Talk

Presented at the UNIONS collaboration weak lensing bi-annual meeting. February 2024 & June 2024, Online.

American Astronomical Society Annual meeting

January 2024, New Orleans, USA.

LISA Collaboration - Invited Talk

LISA collaboration community meeting. January 2023, Online.

Durham-Edinburgh eXtragalactic Workshop XIX

January 2023, Edinburgh, UK.

Probing the Universe with Multimessenger Astrophysics

September 2022, Sestri-Levante, Italy.

UK Cosmology Meeting

May 2022, Newcastle, UK.

Posters

CASCA, GWPAW, SUPA Annual Gathering, Tonale Winter School (received poster prize).

COLLABORATIONS

Euclid Collaboration

Leading a Standard Project on gravitational redshift in the galaxy clusters science working group.

UNIONS Collaboration

Leading a project using UNIONS weak lensing profiles of galaxy clusters for cosmological constraints.

LEADERSHIP AND SERVICE

Large Scale Structure Journal Club

Organised and chaired a weekly cosmology journal club at the University of Edinburgh for over a year.